

Baowen Wei

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EDUCATION

Bachelor of Engineering in Electrical and Biomedical Engineering

Expected Apr 2029

McMaster University — CGPA: 3.8/4.0

Hamilton, ON

SKILLS

Hardware & Electronics: Wire harness assembly, oscilloscopes, function generators, digital multimeters, Raspberry Pi, Analog Discovery Kit

CAD: Altium Designer, SolidWorks, AutoCAD Electrical, Autodesk Inventor

Programming: C, C++, Python, Git/GitHub

Software Tools: Excel, WaveForms, PSpice, MATLAB, Capstone, Figma, BioRender, Microsoft Office, Notion

Professional Skills: Project management, technical communication, creative problem solving, team leadership

TECHNICAL EXPERIENCE

Electrical Engineering Intern

May 2026 – Aug 2026

Westinghouse Electric Company

Kitchener, ON

- Verification and revision of AP1000 power plant auxiliary system wiring diagrams using AutoCAD Electrical.
- Conducted electrical load calculations for existing and proposed warehouse transformer installations using the Ontario Electrical Safety Code (OESC), facility drawings, and panel schedules.
- Prepared the initial revision of a smoke detection system maintenance plan for Bruce Power, updating software, firmware, and manufacturer information.

Powertrain & Wire Harness Team Member

Sept 2025 – Present

McMaster Formula Electric

Hamilton, ON

- Designed electric vehicle wiring architecture, including power distribution, grounding strategy, connector selection, and fusing requirements.
- Developed complete vehicle harness CAD and generated schematics, routing layouts, and manufacturing documentation using Excel and SolidWorks Electrical.
- Researched variable-frequency drive switching strategies and supported motor selection, technical analysis, and documentation

Sensor Data & Signal Processing Team Member

Sept 2025 – Present

McMaster Medical Engineering Design Team

Hamilton, ON

- Implemented image and signal processing algorithms for a tactile sensory substitution device using optical sensors and microcontrollers.
- Conducted requirement-driven research on sensing and processing modules, produced technical reports, and coordinated component selection.

PROJECTS

Automatic Crib Rocker

- Designed and iterated a gear train and barrel cam assembly in Autodesk Inventor to produce controlled oscillatory motion.
- Integrated servo motors, limit switches, and a microcontroller to automate rocking behavior.
- Created manufacturing-ready drawings and optimized 3D-printed components for durability and printability.
- Managed prototyping cycles, validation tests, and mechanical troubleshooting.

Acetabulum Cup Stress Tester

- Developed a Python-based DAQ system on Raspberry Pi for real-time load cell data acquisition and visualization.
- Built breadboard circuitry including signal amplification, filtering, and LED threshold indicators.
- Performed tensile, compression, and 3-point bending tests to evaluate orthopedic implant materials.
- Automated data logging and plotted stress-strain curves for material performance analysis.